

UNITS AND CONSTANTS — CHECK BEFORE ANY NUMBER GOES IN		
QUANTITY	WHAT THE FORMULA NEEDS	THE TRAP
temperature	KELVIN always. K = °C + 273	not 273.15 in this module
gas volume	m³ dm ³ ÷ 1000	1 dm ³ = 1 L = 1000 cm ³ = 0.001 m ³
gas pressure	Pa kPa × 1000	1 atm = 101 325 Pa
solution volume	litres for M = n ÷ V	but mL cancels in M ₁ V ₁ = M ₂ V ₂ and in titration
R	8.314 m ³ Pa K ^{−1} mol ^{−1}	quoted in the question; the only value used
N_A	6.02 × 10 ²³ per mol	
molar volume	24 dm ³ /mol at 20°C, 1 atm	recall only. Calculations go through PV = nRT
entropy against enthalpy	S° tables in J, ΔH in kJ	divide S by 1000 before combining
electrolysis time	seconds for Q = It	convert back to hours only at the end
answers	thermodynamics 2 dp ; elsewhere match the data	state the unit every time

MOLES — EVERYTHING GOES THROUGH THE MIDDLE		
LOOKS LIKE	MOVE	WORKED
mass given	n = m ÷ M	102 ÷ 80.91 = 1.26 mol
moles given, mass wanted	m = n × M	0.80 × 119.00 = 95.20 g
“how many molecules”	N = n × 6.02 × 10 ²³	2.00 mol → 1.20 × 10 ²⁴
molar mass of a formula	count every atom; a number outside a bracket multiplies all of it	Al ₂ (SO ₄) ₃ : the 3 hits S and 4 O = 342.17
unknown element	1. get n from the mole ratio 2. M = mass ÷ n 3. subtract the known atoms	110.98 – 2(35.45) = 40.08 → Ca
before reading any ratio	BALANCE the equation	C ₅ H ₁₂ + 8O ₂ → 5CO ₂ + 6H ₂ O

GASES		
LOOKS LIKE	MOVE	WORKED
moles change, T and P fixed	Avogadro V ₁ /n ₁ = V ₂ /n ₂	5 mol/50 L → 6 mol = 60 L
P and V, T fixed	Boyle P ₁ V ₁ = P ₂ V ₂	3 atm 330 mL → 1 atm 990 mL
V and T, P fixed	Charles V ₁ /T ₁ = V ₂ /T ₂ , T in K	25→70°C = 298→343 K, up 15.10%
anything with n, or a mixture	PV = nRT	n = PV/RT V = nRT/P
P AND T both change	no combined gas law in A105. Take the ratio	V ₂ /V ₁ = (P ₁ /P ₂) (T ₂ /T ₁)
identify a gas from mass + PVT	n = PV/RT, then M = mass ÷ n	0.71/0.0100 = 71 → Cl ₂

SOLUTIONS AND TITRATION		
LOOKS LIKE	MOVE	WORKED
mol/L, molarity, [X]	c = n ÷ V, V in litres	0.100 ÷ 0.250 = 0.400 M
“diluted to”, a flask	c ₁ V ₁ = c ₂ V ₂ DF = how many times weaker	V ₂ is the FLASK size . 0.01 → 0.001 is DF 10
a burette table	1. drop every Rough trial 2. average the accurate titres 3. n = cV on the KNOWN side 4. apply the mole ratio, divide by the other volume	bigger coefficient takes the bigger mole count

SOLUTIONS AND TITRATION		
LOOKS LIKE	MOVE	WORKED
accurate glassware	volumetric flask , bulb pipette, burette	beaker, conical flask and measuring cylinder are approximate only
the other three units	% mass and g/L divide by the whole solution mol/kg divides by the SOLVENT	

LIMITING REAGENT		
LOOKS LIKE	MOVE	WORKED
two amounts given	1. for each reactant, the product it alone gives 2. the smaller one limits	4 CO (1:1)→4, 6 H ₂ (2:1)→3 → H ₂
the classic error	the smaller STARTING amount is not always limiting	Zn 0.0642 vs CuSO ₄ 0.0500 at 1:1 → CuSO ₄
“how much remains”	start – consumed	0.60 – 0.40 = 0.20 mol

ENERGY 1 — HESS AND ENTHALPY		
LOOKS LIKE	MOVE	WORKED
“given that” plus two or three equations	1. write the TARGET equation 2. species already correct? keep 3. wrong side? reverse , flip the sign of ΔH 4. coefficient wrong? scale , multiply ΔH 5. add, cancel, sum the ΔH	reverse and scale are independent +966 reversed then halved = +483
a table of Δ_fH	Σn(products) – Σn(reactants)	an ELEMENT is zero , including H ₂ N ₂ O ₂ Cl ₂ Br ₂ I ₂ reactants first – the opposite way round
a table of bond enthalpies	ΔH = broken (+) + formed (–)	reactants first – the opposite way round
bond counting	each CO ₂ holds 2 C=O, each H ₂ O 2 O–H	3 CO ₂ = 6 bonds, not 3
“gives out 81.67 kJ”	ΔH = –81.67	the minus sign is never written in the question
a combustion that came out positive	the two terms are the wrong way round	combustion is always exothermic

ENERGY 2 — ENTROPY AND FREE ENERGY		
LOOKS LIKE	MOVE	WORKED
“increase or decrease in entropy”	1. count particles : more on the right → up 2. compare states : s < l < aq < g	both tests – one question has equal particles and turns on state alone
a table of S°	ΣS(prod) – ΣS(react)	elements are NOT zero here. O ₂ = 205.0
ΔH and ΔS given, “spontaneous”	1. T to kelvin 2. S in J? divide by 1000 3. ΔG = ΔH – TΔS	<0 spontaneous, >0 not, =0 equilibrium
the double negative	–T × (–ΔS) is plus	–92.2 + 59.30 = –32.90
“at what temperature”	set ΔG = 0, so T = ΔH ÷ ΔS	6.01 ÷ 0.0220 = 273.2 K

SPONTANEITY — THE FOUR CASES		
	ΔH NEGATIVE	ΔH POSITIVE
ΔS positive	spontaneous at ALL T	spontaneous at HIGH T
ΔS negative	spontaneous at LOW T	NEVER spontaneous
why	only TΔS changes with temperature	same signs → temperature cannot flip the verdict

ATOMS, IONS AND ISOTOPES		
LOOKS LIKE	MOVE	WORKED
mass number A, atomic number Z	protons = Z neutrons = A – Z	U-238: 92 p, 146 n
a neutral atom	electrons = protons	

ATOMS, IONS AND ISOTOPES		
LOOKS LIKE	MOVE	WORKED
an ion	positive → lost electrons negative → gained electrons	Ca ²⁺ has 20 p, 18 e
isotopes	same Z, different A	same chemistry, different mass
relative atomic mass from abundances	A _r = Σ(mass × %) ÷ 100	a weighted average, so it is rarely a whole number
electron configuration	fill 1s 2s 2p 3s 3p 4s 3d 4p	4s fills BEFORE 3d, and empties first too
valence electrons	the outermost shell only	Group number gives it for the main groups

BONDING AND POLARITY		
LOOKS LIKE	MOVE	WORKED
metal + non-metal	ionic : electrons transferred	giant lattice, high melting point, conducts when molten or aqueous
non-metal + non-metal	covalent : electrons shared	
Δ(electronegativity)	< 0.5 non-polar covalent 0.5 to 2.0 polar covalent > 2.0 ionic	3.98 – 2.96 = 1.02 → polar covalent
is the MOLECULE polar	polar bonds are not enough — the SHAPE must be asymmetric	CO ₂ has polar bonds and is non-polar ; the dipoles cancel
strongest intermolecular force	H bonded to N, O or F → hydrogen bonding else polar → dipole-dipole else London	more hydrogen bonds → more energy to separate
periodic trends	across : electronegativity up, radius down down : electronegativity down, radius up	F is the most electronegative element

VSEPR — SHAPE FROM THE ELECTRON PAIRS			
DOMAINS PAIRS	LONE	SHAPE	ANGLE
2 0		linear	180°
3 0		trigonal planar	120°
3 1		bent	< 120°
4 0		tetrahedral	109.5°
4 1		trigonal pyramid	107.5°
4 2		bent	104.5°
5 0		trigonal bipyramid	120° and 90°
6 0		octahedral	90°
6 1		square pyramid	< 90°
6 2		square planar	90°
getting there		1. valence electrons on the central atom 2. subtract one per bonded atom 3. what is left ÷ 2 = lone pairs 4. domains = bonds + lone pairs	BrF ₅ : 7 – 5 = 2 → 1 lone pair, 6 domains → square pyramid
the repulsion ORDER, asked as an MCQ	lp-lp > lp-bp > bp-bp		lone pair to lone pair is the strongest ; each extra lone pair squeezes the angle further

KINETICS — 14 MARKS, THE BIGGEST TOPIC ON THE PAPER		
LOOKS LIKE	MOVE	WORKED
“conditions for a successful collision”	CORRECT ORIENTATION and ENOUGH KINETIC ENERGY	both, or it just bounces. Not temperature, not a catalyst
“why does X speed it up”	every reason ends the same way: more successful collisions	surface area, concentration, pressure → more frequent collisions
temperature	the only factor that does both	more collisions and a bigger proportion above E_a
catalyst	alternative route of lower E_a	not used up; k rises; both directions equally
a Maxwell-Boltzmann figure	able to react = area right of E_a	temperature SHIFTS the curve ; concentration only RAISES it; a catalyst moves the E_a LINE
rate = $k[A]^m[B]^n$	orders come from experiment , never the coefficients	exception: an elementary step
overall order	add the powers	$k[X][Y]^2 \rightarrow 3$
“ $[A] \times 4$, what happens to rate”	raise the factor to the power of that order	order 1 $\rightarrow 4^1 = 4$ order 2 and $\times 3 \rightarrow 3^2 = 9$
order from a rates table	find two runs where only one concentration changed	rate same $\rightarrow \theta$ doubles $\rightarrow 1$ $\times 4 \rightarrow 2$
order from a log-log GRAPH	take logs and it becomes a straight line log rate = $n \log[A] + \log k$	gradient = the order intercept = $\log k$, so $k = 10^{\text{intercept}}$
half-life, to find the order	$t_{\frac{1}{2}}$ constant $\rightarrow$ first halving $\rightarrow$ zero doubling $\rightarrow$ second	constant 12 s each time $\rightarrow$ first order
half-life formulas	zero: $t_{\frac{1}{2}} = [A]_0 \div 2k$ first: $t_{\frac{1}{2}} = 0.693 \div k$ second: $t_{\frac{1}{2}} = 1 \div (k[A]_0)$	first order does not depend on concentration; second order is INVERSELY proportional to it
units of k	$M^{(1 - \text{overall order})} s^{-1}$	overall 1 $\rightarrow s^{-1}$ overall 2 $\rightarrow M^{-1}s^{-1}$ overall 3 $\rightarrow M^{-2}s^{-1}$
Arrhenius, one condition	$k = Ae^{-E_a/RT}$, so $E_a = -RT \ln(k \div A)$	$^{\circ}C \rightarrow K$ first. R in kJ if E_a in kJ , and then no conversion is needed
Arrhenius, two conditions	$\ln(k_2/k_1) = E_a/RT_1 - E_a/RT_2$	A cancels when it is unchanged
forward and backward at equilibrium	the two rates are equal , so $k_f[A]^m = k_b[B]^n$	rearrange for the unknown k , then feed it into Arrhenius

EQUILIBRIUM — 12 MARKS, SECOND BIGGEST TOPIC		
LOOKS LIKE	MOVE	WORKED
“write the K_c expression”	1. balance first 2. products over reactants 3. each raised to its coefficient 4. multiply, never add	$CH_4 + 2H_2S \rightleftharpoons CS_2 + 4H_2$ $K_c = [CS_2][H_2]^4 \div ([CH_4][H_2S]^2)$
units of K_c	$M^{\Delta n}$, Δn = powers above – below	$\Delta n = 0 \rightarrow$ no units
add reactant / remove product	shifts right	K unchanged
raise pressure	shifts to fewer moles of GAS	$SO_2Cl_2 \rightleftharpoons SO_2 + Cl_2$ is 1 vs 2, so it goes LEFT and SO_2 falls
raise temperature	shifts the way that absorbs heat	the ONLY change that alters K
catalyst	speeds forward AND reverse EQUALLY	equilibrium arrives sooner, in the same place. K , position and yield all unchanged
heat written in the equation	treat heat as a species	products + heat = exothermic, so heating shifts left
“essentially only product”	$K > 10^2$ product, $< 10^{-2}$ reactant	the boundary is 100

READING A PHASE DIAGRAM — A WHOLE SECTION B QUESTION		
WHAT IS ASKED	WHAT TO DO	WATCH FOR
how many phases	count the labelled regions	solid, liquid, gas $\rightarrow 3$

READING A PHASE DIAGRAM — A WHOLE SECTION B QUESTION		
WHAT IS ASKED	WHAT TO DO	WATCH FOR
boiling point at a pressure	draw a horizontal line at that pressure, find where it crosses the liquid-gas curve, read down	the upper right boundary
melting point at a pressure	same line, but cross the solid-liquid curve	the steep boundary
the triple point	the single point where all three curves meet	read pressure across, temperature down. All three phases coexist
does melting point rise with pressure	look which way the solid-liquid line leans	leans right $\rightarrow$ increases. Water leans LEFT and is the exception – do not answer from memory of water
the decoys	the melting and triple point temperatures are printed close together	iodine: triple point 113.5, melting 113.7, boiling 184

PH AND BUFFERS — IDENTIFY THE CASE, THEN ONE LINE		
LOOKS LIKE	MOVE	WORKED
the four relations	$pH = -\log[H^+]$ $pOH = -\log[OH^-]$ $[H^+] = 10^{-pH}$ $pH + pOH = 14$	
single arrow, a named acid	strong acid: $[H^+] = n \times c$, then $-\log$	monoprotic $n=1$, H_2SO_4 $n=2$
NaOH, KOH, alkaline	strong base: TWO steps. pOH first, then $14 - pOH$	$0.010\ M \rightarrow pOH\ 2.00 \rightarrow pH\ 12.00$ stopping at 2.00 is the trap
double arrow, or K_a , or a %	weak acid: $[H^+] = \% \times c$, then $-\log$	
weak acid plus its own salt	BUFFER: $pH = pK_a + \log(c_{\text{salt}} \div c_{\text{acid}})$	equal concentrations $\rightarrow pH = pK_a$
what a buffer does	resists pH change; the basic half mops up added acid and the acidic half mops up added base	it does not neutralise completely, and diluting barely moves it

OXIDATION STATES — APPLY IN THIS ORDER		
LOOKS LIKE	MOVE	WORKED
uncombined element	0 , and stop	$S_8, O_2, P_4, Na \rightarrow$ all 0
whole species is one atom with a charge	= the charge, and stop	$Fe^{3+} \rightarrow +3$
Group 1, 2, F	+1, +2, –1 , never bend	$Mg(OH)_2 \rightarrow Mg = +2$ immediately
H	+1 , unless bonded to a metal $\rightarrow -1$	NaH, CaH_2
O	–2 , unless peroxide $\rightarrow -1$	H_2O_2 and $O_2^{2-} \rightarrow 0 = -1$
Cl, Br, I	-1 in a binary compound. If O or F is also there, do NOT assign	$HClO_4 \rightarrow Cl$ is the unknown
whatever is left	$\Sigma(\text{count} \times \text{state}) = \text{total charge}$	neutral $\rightarrow 0$ polyatomic ion $\rightarrow$ its charge
a fast self-check	a state above +7 means a charge was forgotten	MnO_4^- summed to 0 gives +8, which is impossible

REDOX AND CELLS		
LOOKS LIKE	MOVE	WORKED
“what is oxidised / reduced”	1. assign states on both sides 2. find the atoms whose number changed 3. up = oxidised, down = reduced	nothing changed $\rightarrow$ not a redox reaction
“the oxidising agent”	the thing that gets REDUCED	the two words are always opposites
balance in acid	1. split into two half-equations 2. balance all but O and H 3. add H_2O for O, then H^+ for H 4. add electrons to balance charge 5. scale so the electrons cancel, then add	check: same atoms and same total charge each side
balance in base	do all of the above via H^+ first , then add the same number of OH^- to both sides	starting from OH^- is much harder

REDOX AND CELLS		
LOOKS LIKE	MOVE	WORKED
galvanic or electrolytic	no battery, spontaneous, E° positive $\rightarrow$ galvanic power source, driven, E° negative $\rightarrow$ electrolytic	
what NEVER changes	anode = oxidation, cathode = reduction, in BOTH	only the signs swap
the signs	galvanic: anode $-$, cathode $+$	electrolytic: anode $+$, cathode $-$
which one is reduced	the more positive E°	-0.28 beats -0.74
“calculate E°_{cell} ”	$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{ox}} + E^{\circ}_{\text{red}}$	reverse the oxidised half and flip its sign
a scaled half-equation	E° is NOT multiplied. Volts are per electron	scale the equation, leave $+0.80$ alone
electrons flow	always anode to cathode	through the external wire

ELECTROLYSIS — THE CHAIN		
LOOKS LIKE	MOVE	WORKED
mass from current and time	1. $Q = I \times t$, t in seconds 2. $n(e^-) = Q \div 96500$ 3. divide by the electrons in the half-equation 4. $\times M$ for grams	2 A for 1 h $\rightarrow Q = 7200 \rightarrow 0.0746\ \text{mol } e^-$ $Na^+ + e^-$ is 1:1 $\rightarrow 1.72\ \text{g}$
“how long” or “what current”	run the chain backwards	$\text{mol} \times 96500 \div Q \rightarrow s \rightarrow \div 3600$
plating or refining	the object is the CATHODE	metal always deposits at the cathode
if the question prints an E° value	use the question, not the table	the tables disagree on Ni (-0.25 vs -0.28)

EVERY FORMULA, IN ONE PLACE		
moles	$n = m \div M$ $N = n N_A$ $c = n \div V$	
gas	$PV = nRT$ $P_1V_1 = P_2V_2$ $V_1/T_1 = V_2/T_2$ $V_1/n_1 = V_2/n_2$	
dilute	$c_1V_1 = c_2V_2$ $DF = V_{\text{final}} \div V_{\text{initial}}$	
acid	$pH = -\log[H^+]$ $[H^+] = 10^{-pH}$ $pH + pOH = 14$	
buffer	$pH = pK_a + \log(c_{\text{salt}} \div c_{\text{acid}})$ $pK_a = -\log K_a$	
eqm	$K_c = [\text{products}] \div [\text{reactants}]$, each to its coefficient units $M^{\Delta n}$	
ΔH	$\Sigma n\Delta_f H(\text{prod}) - \Sigma n\Delta_f H(\text{react})$ or bonds broken (+) + bonds formed (–)	
ΔS	$\Sigma S^{\circ}(\text{prod}) - \Sigma S^{\circ}(\text{react})$ S in J, so $\div 1000$ first	
ΔG	$\Delta G = \Delta H - T\Delta S$ crossover at $T = \Delta H \div \Delta S$	
rate	$\text{rate} = k[A]^m[B]^n$ units $M^{(1-\text{order})}s^{-1}$	
Arrh	$k = Ae^{-E_a/RT}$ $\ln(k_2/k_1) = E_a/RT_1 - E_a/RT_2$	
cell	$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{ox}} + E^{\circ}_{\text{red}}$ positive means spontaneous	
elec	$Q = It$ $n(e^-) = Q \div 96500$ then $\div$ the electrons per ion, then $\times M$	
temp	$K = ^{\circ}C + 273$ every single time	

WHAT EACH SYMBOL MEANS		
n	amount, in moles	K_a acid dissociation constant. Bigger means stronger
m	mass, in grams	
M	means TWO things — molar mass in g/mol, and molarity in mol/L. Read which from the units	ΔH enthalpy. Negative = exothermic, heat out
c	concentration, mol/L. Used here for molarity to keep it apart from molar mass	ΔS entropy, disorder. Positive = more disordered
[X]	concentration of X, mol/L	ΔG free energy. Negative = spontaneous
V	volume — m^3 in PV = nRT, litres in $c = n/V$	$\Delta_f H$ enthalpy of formation . Zero for an element
P	pressure, in Pa	E_a activation energy, the hill to climb
T	temperature, in kelvin , always	k rate constant. A frequency factor
R	gas constant, 8.314	Q charge in coulombs. I current in amps
N_A	Avogadro, 6.02×10^{23} per mol	F Faraday, 96 500 C per mole of electrons
K_c	equilibrium constant — only temperature changes it	E° standard electrode potential, in volts
Q_c	the same expression, but at any instant	$\rightleftharpoons$ reversible. $\rightarrow$ goes to completion
		p anything = –log of it , so pH, pOH and pK_a all work the same way